SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0328	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Keerthika S	**Reg. No.:**	192371013
Section / Batch:	2023-2027	**Date:**	23.10.26

Code of Conduct

I Keerthika S 192371013 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Keerthika S

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

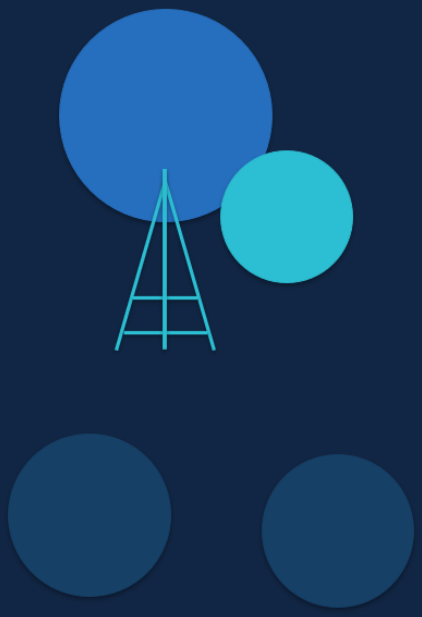
Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Cell Structure and Cellular concept

MOBILE COMPUTING • PRESENTATION

Cell Structure and Cellular Concept

How mobile networks divide areas into cells to provide coverage.

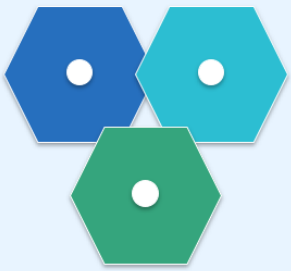


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What Is the Cellular Concept?

The basic idea behind mobile network coverage

- A large area is divided into smaller areas called cells.
- Each cell is served by a base station or cell tower.
- Your phone connects to a nearby tower using radio signals.
- As you move, your phone can connect to another cell.
- This system helps provide coverage to many users.



Each cell covers part of the area.

Cell and Base Station

Two important parts of a cellular network



A cell

A small geographic area served by a base station. Many cells work together to cover a town or city.



Base station / cell tower

It sends and receives radio signals between mobile phones and the network.

Why Do We Divide an Area into Cells?

Benefits of the cellular concept

Better coverage

Cells help cover large areas, including towns and cities.

More users

Different cells can serve many people at the same time.

Frequency reuse

Frequencies can be reused in cells far enough apart.

Easy expansion

More cells can be added where more people need service.

Cells help networks use limited radio resources more efficiently.

How It Works When You Use Your Phone

A simple example: making a call while moving

1

Phone

Your phone sends a radio signal.

2

Cell tower

A nearby tower receives the signal.

3

Network

The network routes your call or data.

4

Next cell

If you move, your connection can transfer to another cell.

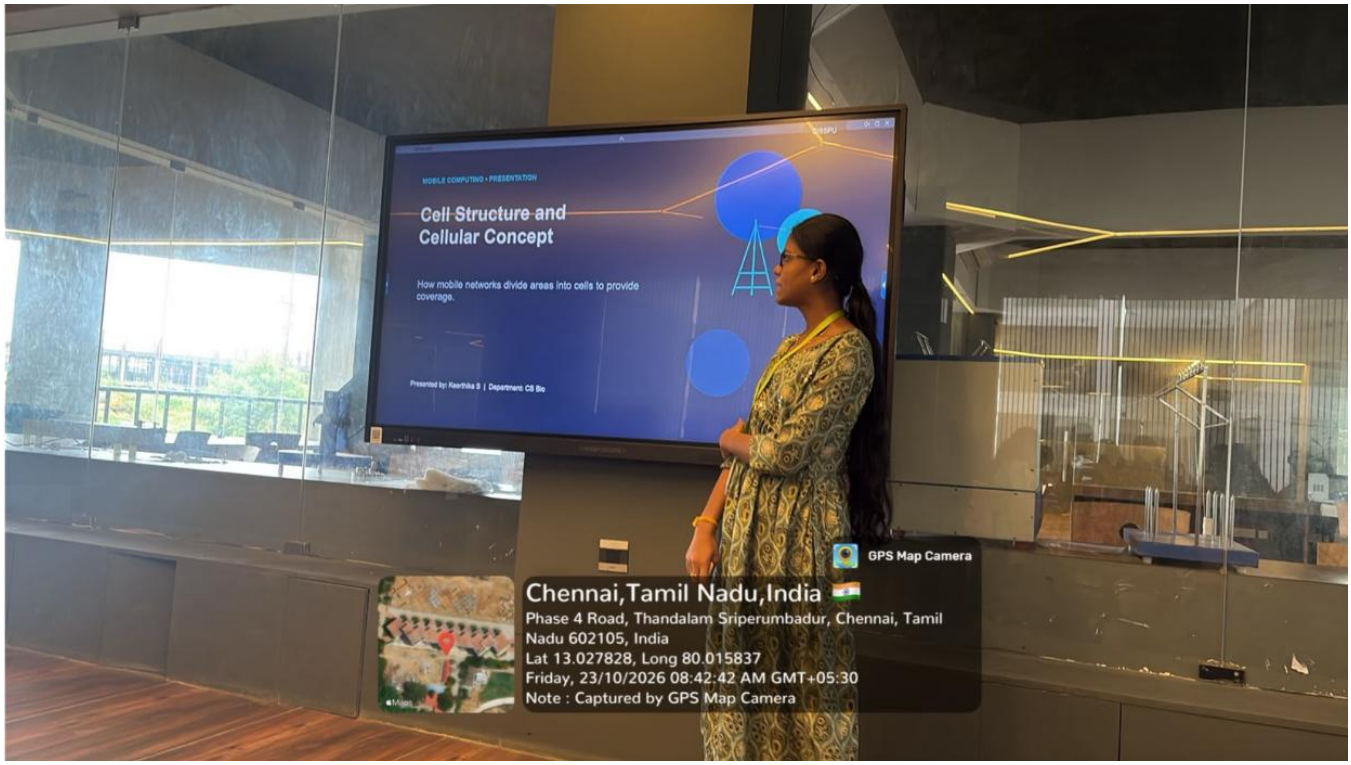
This transfer between cells is called handoff or handover.

Summary and Conclusion

- A cellular network divides a large area into smaller cells.
- Each cell is served by a base station or cell tower.
- Cells improve coverage and help support many users.
- Frequency reuse helps use radio resources efficiently.
- Handoff keeps a connection working as a user moves.



Thank You!


MOBILE COMPUTING • PRESENTATION
Cell Structure and Cellular Concept
How mobile networks divide areas into cells to provide coverage.
Presented by: Keerthika S | Department: CS Bio
GPS Map Camera
Chennai,Tamil Nadu,India
Phase 4 Road, Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India
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